Urban Policies to Combat Climate Change: Assessing the Impact of Congestion Pricing and Low-Emission Zones using Machine Learning Techniques

Ferdinand Loser*

July 22, 2026

Course: *Machine Learning in Econometrics*
Supervisor: Prof. Daniel Wilhelm

Ludwig-Maximilian University of Munich, M. Sc. Economics Program

Executive Summary

Abstract content goes here. This section should provide a concise summary of the research question, methodology, key findings, and implications of the study. It should be clear and informative, allowing readers to quickly understand the purpose and results of the paper.

*Ludwig-Maximilian University of Munich, ferdinand.loser@campus.lmu.de
Replication files are available in the GitHub repository for this project.

1 Introduction

2 Theoretical Background

We define our Partially Linear Model for a given city i at time t as follows:

$$Y_{it} = \theta_0 D_{it} + g_0(X_{it}) + U_{it}, \quad E[U_{it}|D_{it}, X_{it}] = 0$$

$$D_{it} = m_0(X_{it}) + V_{it}, \quad E[V_{it}|X_{it}] = 0$$

Where:

- Y_{it} is the natural logarithm of transport CO_2 emissions.
- D_{it} is the policy treatment indicator (e.g., Congestion Pricing).
- X_{it} is the high-dimensional matrix of city-level covariates.
- θ_0 is the true causal parameter of interest.
- $g_0(X_{it})$ is the true expected baseline emissions given city characteristics.
- $m_0(X_{it})$ is the true propensity score (the probability of policy adoption).
- U_{it} and V_{it} are the respective error terms.

Our nuisance parameters are defined as $\eta = (g, m)$, with their true values being $\eta_0 = (g_0, m_0)$.

2.1 The Orthogonal Score Function

To estimate θ_0 without regularization bias from the machine learning algorithms, we rely on the Robinson (1988) partialling-out approach. We define the moment condition (score function) ψ for a single observation $W = (Y, D, X)$ as:

$$\psi(W; \theta, \eta) = (Y - \theta D - g(X))(D - m(X))$$

The true parameter θ_0 uniquely solves the population moment condition $E[\psi(W; \theta_0, \eta_0)] = 0$.

2.2 Proof of Neyman Orthogonality

Neyman orthogonality requires that the moment condition is locally insensitive to small estimation errors in the machine learning nuisance functions $\hat{\eta} = (\hat{g}, \hat{m})$.

Mathematically, the Gateaux derivative of the expected score with respect to the nuisance parameter η, evaluated at the true parameter η_0, must equal zero in all directions.

Let us perturb the true nuisance parameters by a scalar r in an arbitrary direction $\tilde{\eta} = (\tilde{g}, \tilde{m})$, such that $\eta = \eta_0 + r\tilde{\eta}$. We evaluate the expectation of the score function under this perturbed state:

$$E[\psi(W; \theta_0, \eta_0 + r\tilde{\eta})] = E[(Y - \theta_0 D - (g_0(X) + r\tilde{g}(X)))(D - (m_0(X) + r\tilde{m}(X)))]$$

Substitute $U = Y - \theta_0 D - g_0(X)$ and $V = D - m_0(X)$ into the equation:

$$E[\psi(W; \theta_0, \eta_0 + r\tilde{\eta})] = E[(U - r\tilde{g}(X))(V - r\tilde{m}(X))]$$

Expanding the product yields:

$$E[\psi(W; \theta_0, \eta_0 + r\tilde{\eta})] = E[UV] - rE[U\tilde{m}(X)] - rE[V\tilde{g}(X)] + r^2 E[\tilde{g}(X)\tilde{m}(X)]$$

By the definitions of our DGP, $E[U|X, D] = 0$ implies that $E[U|X] = 0$ and $E[UV] = 0$. Similarly, $E[V|X] = 0$. Applying the Law of Iterated Expectations, the linear perturbation terms vanish:

$$E[U\tilde{m}(X)] = E[E[U|X]\tilde{m}(X)] = 0$$

$$E[V\tilde{g}(X)] = E[E[V|X]\tilde{g}(X)] = 0$$

This leaves only the second-order term:

$$E[\psi(W; \theta_0, \eta_0 + r\tilde{\eta})] = r^2 E[\tilde{g}(X)\tilde{m}(X)]$$

Taking the Gateaux derivative with respect to r and evaluating at $r = 0$:

$$\frac{\partial}{\partial r} E[\psi(W; \theta_0, \eta_0 + r\tilde{\eta})]\bigg|_{r=0} = \frac{\partial}{\partial r}\left(r^2 E[\tilde{g}(X)\tilde{m}(X)]\right)\bigg|_{r=0} = 2rE[\tilde{g}(X)\tilde{m}(X)]\bigg|_{r=0} = 0$$

$\blacksquare$

3 Results

	Sprawling Regional Hubs	Dense Progressive Metropolises
National Climate Pact Participation	0.15	0.26
Number of public libraries	1.50	2.29
Temperature Anomaly	0.34	0.24
Indicator: coastal city (1=yes)	0.39	0.28
Share of employment in services	0.22	0.30
Share of employment in public sector	0.27	0.20
Share of employment in logistics	0.28	0.23
Public transport quality index (0-100)	52.71	62.91
Log Population per km^2	6.64	7.91
Share of employment in manufacturing	0.24	0.28

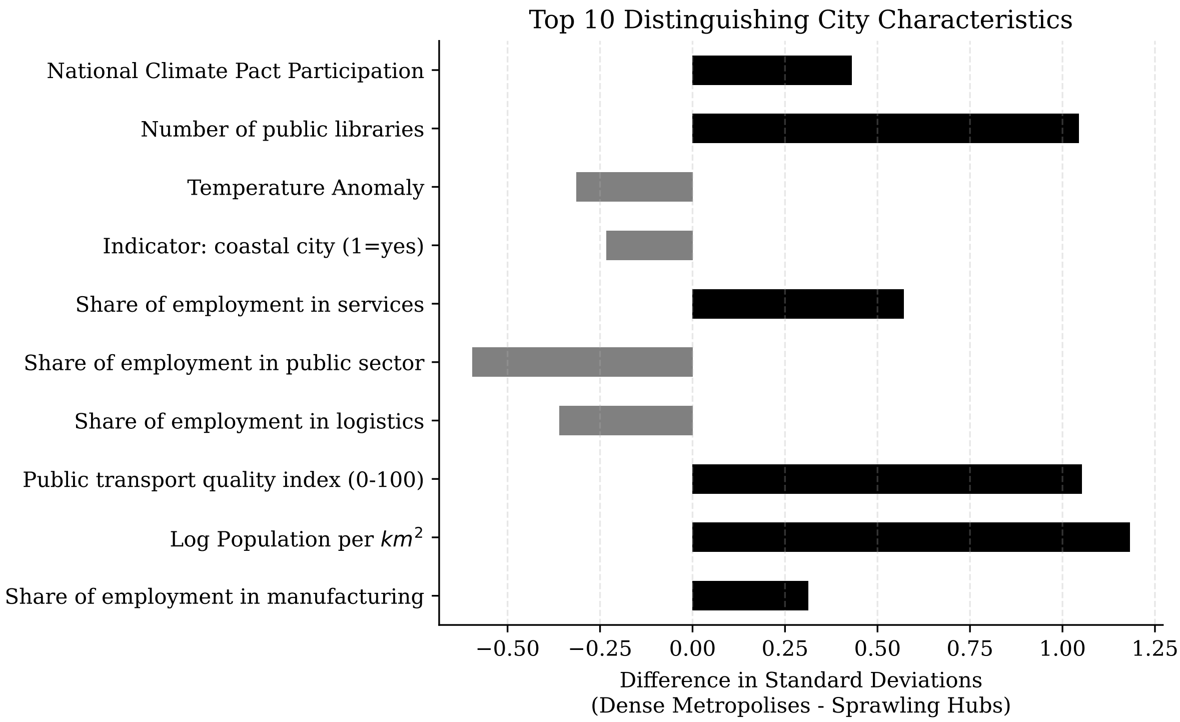

Figure 1: Cluster Differences.

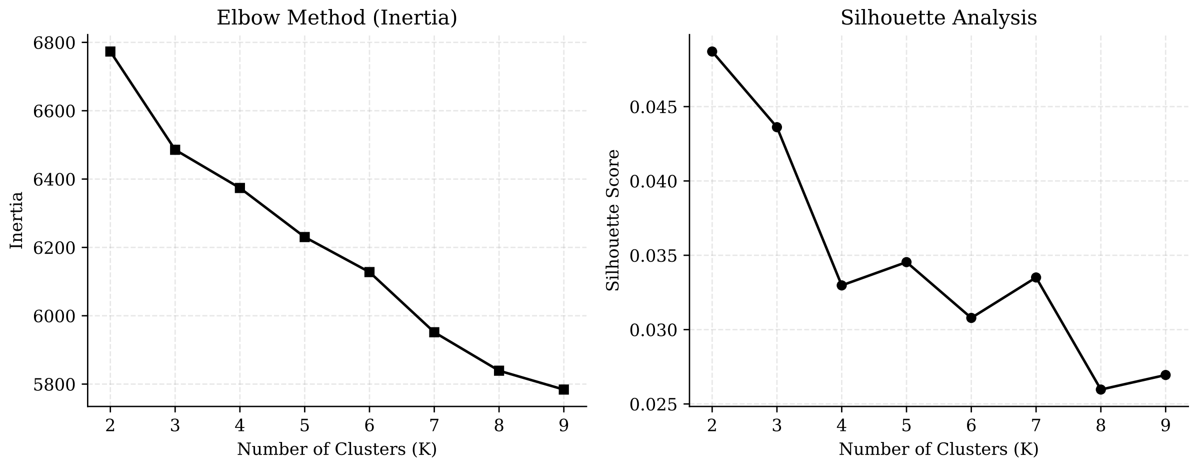

Figure 2: K-Means Evaluation.

Table 1: Double ML Causal Estimates on Log Transport CO_2 Emissions

Variable	**Model 1: Average Effects** *(Main & Synergy)*	**Model 2: Heterogeneity** *(by City Type)*
Congestion Pricing (CP)	-0.41***	-0.54***
	(0.07)	(0.08)
Low-Emission Zone (LEZ)	-0.15**	-0.30***
	(0.08)	(0.08)
CP × LEZ (Synergy)	-0.79***	
	(0.19)	
CP × Dense Metropolis		-0.76***
		(0.14)
LEZ × Dense Metropolis		-0.51***
		(0.17)
ML Learner	Random Forest	Random Forest
Cross-Fitting Folds	5	5
Observations	3,111	3,111

Notes: Standard errors clustered at the city level in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. The dependent variable is log transport CO_2.

Table 2: Sensitivity Analysis: Penalized Linear Models

Variable	**Model 1: Average Effects** *(Main & Synergy)*	**Model 2: Heterogeneity** *(by City Type)*
Congestion Pricing (CP)	-0.33***	-0.37***
	(0.08)	(0.08)
Low-Emission Zone (LEZ)	0.06	0.01
	(0.07)	(0.08)
CP × LEZ (Synergy)	-1.05***	
	(0.29)	
CP × Dense Metropolis		0.02
		(0.16)
LEZ × Dense Metropolis		-0.37
		(0.28)
ML Learner	Lasso / Logistic CV	Lasso / Logistic CV
Cross-Fitting Folds	5	5
Observations	3,111	3,111

Notes: Standard errors clustered at the city level in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. The dependent variable is log transport CO_2.

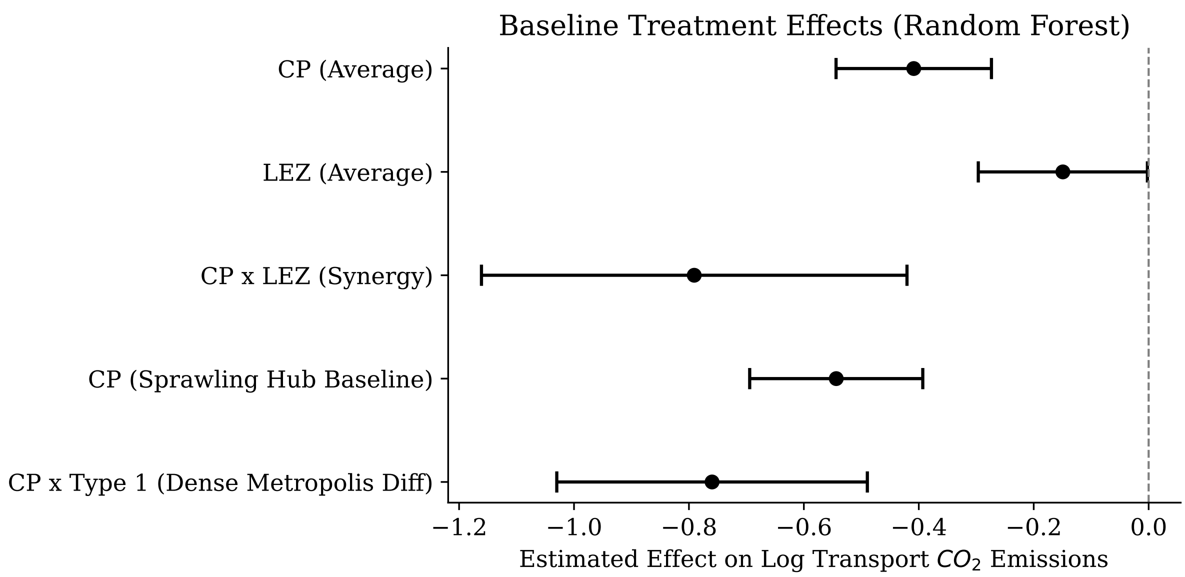

Figure 3: Baseline: Treatment Effects.

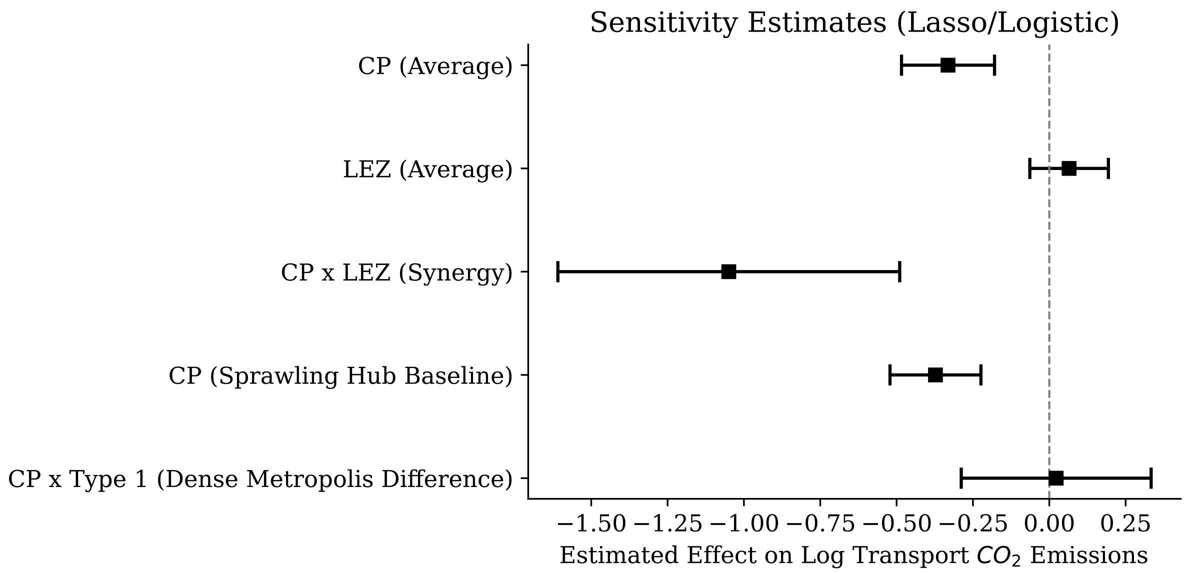

Figure 4: Sensitivity: Treatment Effects.

4 Conclusion

Appendix

Declaration of Autorship

I confirm that this report is based on my own work. In preparing this report, I have not received any help from another human nor have I discussed any aspects of the empirical project with others.

Ferdinand Loser